

InternNova

Week 2 Assignment

NumPy and Pandas for Data Analytics

Task 1: NumPy Arrays

In this task, I created a NumPy array containing numerical values and performed basic array operations. I displayed the array and checked its shape, size, and data type to understand its structure. This task helped me learn how NumPy arrays store and organize numerical data for efficient calculations in Python.

Output:

```
Shell x
>>> %Run -c $EDITOR_CONTENT
Original Array
[11 22 33 44 55 66 77 88 99 61]

Shape: (10,)

Size: 10

Datatype: int64

One Dimensional Array: [10 20 30 40 50]

Two Dimensional Array: [[10 20 30]
 [50 60 70]]

Two Dimensional Array Shape: (2, 3)

>>>
```

Task 2: NumPy Indexing, Slicing & Reshaping

In this task, I performed indexing, slicing, and reshaping operations on NumPy arrays. Indexing helped me access individual elements, slicing allowed me to extract a specific portion of the array, and reshaping helped me convert a one-dimensional array into a two-dimensional array. This task improved my understanding of how to access and organize array data.

Output:

```
Shell >
>>> %Run -c $EDITOR_CONTENT
Original Array: [ 10  20  30  40  50  60  70  80  90 100]

Indexing
First Element: 10
Third Element: 30
Last Element: 100

Slicing
Elements From Index 1 to 3: [20 30 40]
First Three Elements: [10 20 30]

Two Dimensional Array: [[1 2 3]
 [4 5 6]
 [7 8 9]]

Accessing Rows & Columns

First Row [1 2 3]
Second Row [4 5 6]
First Column [1 4 7]

Element in Row 1 & Column 2 6

Original 1D Array [1 2 3 4 5 6]
Reshaped 2D Array
[[1 2 3]
 [4 5 6]]
```

Task 3: NumPy Mathematical & Statistical Operations

In this task, I performed mathematical and statistical operations using NumPy, including addition, subtraction, multiplication, division, mean, median, minimum, maximum, standard deviation, and sum. These operations helped me understand how to calculate and summarize numerical data. This task improved my knowledge of using NumPy for data analysis and statistical calculations.

Output:

```
>>> %Run -c $EDITOR_CONTENT
Array A: [10 20 30 40 50]
Array B: [ 2  4  5  8 10]

--- Mathematical Operations ---
Addition: [12 24 35 48 60]
Subtraction: [ 8 16 25 32 40]
Multiplication: [ 20  80 150 320 500]
Division: [5. 5. 6. 5. 5.]

--- Statistical Functions ---
Mean: 30.0
Median: 30.0
Minimum: 10
Maximum: 50
Standard Deviation: 14.142135623730951
Sum: 150

>>>
```

Task 4: Pandas Series & DataFrame

In this task, I created a Pandas Series and a DataFrame containing information such as employee names, ages, and departments. I displayed the DataFrame, its index, and column names to understand how data is organized in rows and columns. This task helped me learn the basics of Pandas, which is widely used for data manipulation and analysis.

Output:

```
Shell ×
Pandas Series:
0      85
1      90
2      78
3      92
4      88
dtype: int64

Series with Student Names:
Tejaswi      85
Anu          90
Priya        78
dtype: int64
Pandas DataFrame:
   Name  Age  Marks
0  Tejaswi   20    85
1    Anu    21    90
2  Priya    20    78

DataFrame Columns: Index(['Name', 'Age', 'Marks'], dtype='str')

DataFrame Shape: (3, 3)
<class 'pandas.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Name    3 non-null      str
1   Age     3 non-null      int64
2   Marks   3 non-null      int64
dtypes: int64(2), str(1)
memory usage: 204.0 bytes

DataFrame Information: None
```

Task 5: Reading & Inspecting Data

In this task, I loaded a CSV dataset using Pandas and performed data inspection operations. I displayed the first and last five rows, checked the number of rows and columns, examined column names and data types, and identified missing values. This task helped me understand the structure and quality of a dataset before performing further analysis.

Output:

--- First 5 Rows ---

	Student_ID	Name	Age	Marks	Department	City	Attendance \
0	1001	Tejaswi	18	90	IT	Hyderabad	92.21
1	1002	Anu	23	55	CIVIL	Tirupati	99.01
2	1003	Priya	22	89	IT	Vijayawada	82.53
3	1004	Ravi	21	45	MECH	Nellore	70.04
4	1005	Kiran	21	88	IT	Visakhapatnam	65.49

Monthly_Stipend

0	17326
1	20580
2	19781
3	7691
4	21521

--- Last 5 Rows ---

	Student_ID	Name	Age	Marks	Department	City	Attendance \
95	1096	Harsha	20	49	EEE	Nellore	96.59
96	1097	Swathi	19	63	CIVIL	Vijayawada	99.28
97	1098	Varun	22	51	IT	Guntur	93.07
98	1099	Bhavya	22	63	ECE	Guntur	92.28
99	1100	Sandeep	18	97	CSE	Guntur	87.49

Monthly_Stipend

95	10067
96	19437
97	16802
98	6002
99	6901

--- Shape ---

(100, 8)

--- Column Names ---

```
Index(['Student_ID', 'Name', 'Age', 'Marks', 'Department', 'City',  
      'Attendance', 'Monthly_Stipend'],  
      dtype='object')
```

--- Data Types ---

```
Student_ID    int64
Name          object
Age           int64
Marks         int64
Department    object
City          object
Attendance    float64
Monthly_Stipend int64
dtype: object
```

--- Dataset Information ---

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 100 entries, 0 to 99
```

```
Data columns (total 8 columns):
```

```
# Column      Non-Null Count  Dtype
```

```
---  ---
0 Student_ID    100 non-null  int64
1 Name          100 non-null  object
2 Age           100 non-null  int64
3 Marks         100 non-null  int64
4 Department    100 non-null  object
5 City          100 non-null  object
6 Attendance    100 non-null  float64
7 Monthly_Stipend 100 non-null  int64
dtypes: float64(1), int64(4), object(3)
memory usage: 6.4+ KB
```

--- Statistical Summary ---

	Student_ID	Age	Marks	Attendance	Monthly_Stipend
count	100.000000	100.000000	100.000000	100.000000	100.000000
mean	1050.500000	21.190000	71.470000	81.528000	15724.870000
std	29.011492	1.910603	15.24378	10.731613	5540.901886
min	1001.000000	18.000000	45.000000	65.490000	5702.000000
25%	1025.750000	20.000000	60.000000	70.687500	11330.000000
50%	1050.500000	21.000000	72.000000	81.730000	15891.000000
75%	1075.250000	23.000000	83.250000	92.227500	20449.500000
max	1100.000000	24.000000	99.000000	99.280000	24934.000000

Task 6: Selecting, Filtering & Sorting Data

In this task, I selected specific columns and rows from a Pandas DataFrame, filtered records based on conditions, and sorted data in ascending and descending order. For example, I filtered students who scored 80 or above and sorted students according to their marks. This task helped me extract relevant information and organize data for better analysis.

Output:

```

--- Marks in Ascending Order ---
...
Student_ID   Name   Age   Marks   Department   City   Attendance \
3            1004   Ravi   21      45          MECH   Nellore   70.04
37           1038   Varun  23      46          IT     Guntur   92.47
46           1047   Arjun  21      46          MECH   Nellore  86.11
69           1070   Lakshmi 21      46          EEE    Tirupati 65.74
16           1017   Swathi 21      47          CSE    Tirupati 69.59
..           ...     ...    ...     ...         ...     ...
94           1095   Keerthi 24      97          EEE    Tirupati 94.53
99           1100   Sandeep 18      97          CSE    Guntur   87.49
60           1061   Tejaswi 24      99          MECH   Nellore  70.12
40           1041   Tejaswi 19      99          CSE    Visakhapatnam 94.03
44           1045   Kiran  18      99          EEE    Nellore  86.21

```

```

Monthly_Stipend
3            7691
37           10937
46           14210
69           7642
16           9996
..           ...
94           6802
99           6901
60           8534
40           18417
44           20406

```

[100 rows x 8 columns]

```

--- Marks in Descending Order ---
Student_ID   Name   Age   Marks   Department   City   Attendance \
44           1045   Kiran  18      99          EEE    Nellore  86.21
60           1061   Tejaswi 24      99          MECH   Nellore  70.12
40           1041   Tejaswi 19      99          CSE    Visakhapatnam 94.03
94           1095   Keerthi 24      97          EEE    Tirupati 94.53
99           1100   Sandeep 18      97          CSE    Guntur   87.49
..           ...     ...    ...     ...         ...     ...
16           1017   Swathi 21      47          CSE    Tirupati 69.59
37           1038   Varun  23      46          IT     Guntur   92.47
46           1047   Arjun  21      46          MECH   Nellore  86.11
69           1070   Lakshmi 21      46          EEE    Tirupati 65.74
3            1004   Ravi   21      45          MECH   Nellore  70.04

```

```

--- Marks in Descending Order ---
Student_ID   Name   Age   Marks   Department   City   Attendance \
44           1045   Kiran  18      99          EEE    Nellore  86.21
60           1061   Tejaswi 24      99          MECH   Nellore  70.12
40           1041   Tejaswi 19      99          CSE    Visakhapatnam 94.03
94           1095   Keerthi 24      97          EEE    Tirupati 94.53
99           1100   Sandeep 18      97          CSE    Guntur   87.49
..           ...     ...    ...     ...         ...     ...
16           1017   Swathi 21      47          CSE    Tirupati 69.59
37           1038   Varun  23      46          IT     Guntur   92.47
46           1047   Arjun  21      46          MECH   Nellore  86.11
69           1070   Lakshmi 21      46          EEE    Tirupati 65.74
3            1004   Ravi   21      45          MECH   Nellore  70.04

```

```

Monthly_Stipend
44           20406
60           8534
40           18417
94           6802
99           6901
..           ...
16           9996
37           10937
46           14210
69           7642
3            7691

```

[100 rows x 8 columns]

```

--- Original DataFrame ---
***
  Student_ID  Name  Age  Marks  Department  City  Attendance  \
0      1001  Tejaswi  18    90         IT   Hyderabad    92.21
1      1002    Anu   23    55        CIVIL   Tirupati    99.01
2      1003   Priya  22    89         IT   Vijayawada    82.53
3      1004    Ravi  21    45        MECH    Nellore    70.04
4      1005   Kiran  21    88         IT  Visakhapatnam    65.49
..      ...     ...   ...   ...     ...     ...
95     1096  Harsha  20    49         EEE    Nellore    96.59
96     1097  Swathi  19    63        CIVIL   Vijayawada    99.28
97     1098   Varun  22    51         IT     Guntur    93.07
98     1099  Bhavya  22    63         ECE     Guntur    92.28
99     1100  Sandeep  18    97         CSE     Guntur    87.49

```

```

      Monthly_Stipend
0          17326
1          20580
2          19781
3           7691
4          21521
..           ...
95          10067
96          19437
97          16802
98           6002
99           6901

```

[100 rows x 8 columns]

```

--- Selected Columns ---
      Name  Marks
0  Tejaswi    90
1    Anu    55
2   Priya    89
3    Ravi    45
4   Kiran    88
..     ...    ...
95  Harsha    49
96  Swathi    63
97   Varun    51
98  Bhavya    63
99  Sandeep    97

```

[100 rows x 2 columns]

```

--- First Three Rows ---
  Student_ID  Name  Age  Marks  Department  City  Attendance  \
0      1001  Tejaswi  18    90         IT   Hyderabad    92.21
1      1002    Anu   23    55        CIVIL   Tirupati    99.01
2      1003   Priya  22    89         IT   Vijayawada    82.53

      Monthly_Stipend
0          17326
1          20580
2          19781

```

--- Students with Marks Greater Than 80 ---

	Student_ID	Name	Age	Marks	Department	City	Attendance	\	Monthly_Stipend
0	1001	Tejaswi	18	90	IT	Hyderabad	92.21	0	17326
2	1003	Priya	22	89	IT	Vijayawada	82.53	2	19781
4	1005	Kiran	21	88	IT	Visakhapatnam	65.49	4	21521
5	1006	Sneha	24	88	ECE	Vijayawada	73.04	5	15721
6	1007	Arjun	18	87	MECH	Guntur	69.61	6	11461
7	1008	Divya	22	81	CIVIL	Visakhapatnam	88.72	7	15284
9	1010	Lakshmi	18	83	MECH	Vijayawada	82.72	9	22152
11	1012	Pooja	24	87	MECH	Tirupati	85.34	11	14256
21	1022	Anu	20	81	IT	Vijayawada	82.10	21	14558
24	1025	Kiran	23	92	MECH	Nellore	75.03	24	17084
27	1028	Divya	23	87	EEE	Vijayawada	84.43	27	12350
34	1035	Keerthi	24	88	IT	Hyderabad	97.66	34	8658
40	1041	Tejaswi	19	99	CSE	Visakhapatnam	94.03	40	18417
44	1045	Kiran	18	99	EEE	Nellore	86.21	44	20406
45	1046	Sneha	24	91	CSE	Guntur	95.37	45	11567
48	1049	Manoj	22	90	MECH	Guntur	78.10	48	24210
50	1051	Sai	23	92	ECE	Visakhapatnam	87.82	50	14785
52	1053	Rahul	20	95	ECE	Guntur	80.89	52	9047
55	1056	Harsha	18	81	CSE	Nellore	91.11	55	15654
59	1060	Sandeep	22	88	EEE	Tirupati	85.81	59	10763
60	1061	Tejaswi	24	99	MECH	Nellore	70.12	60	8534
61	1062	Anu	23	81	ECE	Nellore	93.86	61	19698
65	1066	Sneha	20	89	EEE	Visakhapatnam	70.79	65	18896
72	1073	Rahul	21	87	CSE	Hyderabad	66.87	72	17725
73	1074	Nikhil	18	84	CIVIL	Hyderabad	85.69	73	6902
74	1075	Keerthi	22	83	IT	Nellore	88.83	74	17438
76	1077	Swathi	20	84	IT	Tirupati	76.13	76	16616
78	1079	Bhavya	21	94	CSE	Hyderabad	95.63	78	15465
80	1081	Tejaswi	24	96	ECE	Vijayawada	66.52	80	19469
83	1084	Ravi	23	83	IT	Guntur	73.73	83	24178
90	1091	Sai	21	82	MECH	Tirupati	94.17	90	24934
94	1095	Keerthi	24	97	EEE	Tirupati	94.53	94	6802
99	1100	Sandeep	18	97	CSE	Guntur	87.49	99	6901

--- CSE Students with Marks Greater Than 80 ---

Student_ID	Name	Age	Marks	Department	City	Attendance	\
40	1041	Tejaswi	19	99	CSE	Visakhapatnam	94.03
45	1046	Sneha	24	91	CSE	Guntur	95.37
55	1056	Harsha	18	81	CSE	Nellore	91.11
72	1073	Rahul	21	87	CSE	Hyderabad	66.87
78	1079	Bhavya	21	94	CSE	Hyderabad	95.63
99	1100	Sandeep	18	97	CSE	Guntur	87.49

Monthly_Stipend

40	18417
45	11567
55	15654
72	17725
78	15465
99	6901

Task 7: Handling Missing Values

In this task, I identified and counted missing values in a dataset using Pandas functions such as `isnull()` and `isna()`. I learned how to remove rows containing missing values and fill empty values using suitable methods, such as the mean. Handling missing data is important because it improves data quality and helps produce more reliable analysis results.

Output:

```

** --- Original DataFrame ---
  Student_ID  Name  Age  Marks  Department  City  Attendance \
0      1001  Tejaswi  18.0  90.0      IT      Hyderabad      92.21
1      1002    Anu   23.0  55.0     CIVIL      Tirupati      99.01
2      1003   Priya  22.0  89.0      IT      Vijayawada      NaN
3      1004    Ravi   NaN  45.0     MECH      Nellore      70.04
4      1005   Kiran  21.0  88.0      IT  Visakhapatnam      65.49
..      ...      ...      ...      ...      ...      ...
95     1096   Harsha  20.0  49.0      EEE      Nellore      96.59
96     1097   Swathi  19.0  63.0     CIVIL  Vijayawada      99.28
97     1098   Varun  22.0  51.0      IT      Guntur      93.07
98     1099   Bhavya  22.0  63.0      ECE      Guntur      92.28
99     1100  Sandeep  18.0  97.0      CSE      Guntur      87.49

```

```

  Monthly_Stipend
0      17326.0
1      20580.0
2      19781.0
3       7691.0
4      21521.0
..      ...
95     10067.0
96     19437.0
97     16802.0
98       6002.0
99     6901.0

```

[100 rows x 8 columns]

```

--- Missing Values (True/False) ---
  Student_ID  Name  Age  Marks  Department  City  Attendance \
*** 0      False  False  False  False      False  False      False
1      False  False  False  False      False  False      False
2      False  False  False  False      False  False      True
3      False  False  True   False      False  False      False
4      False  False  False  False      False  False      False
..      ...      ...      ...      ...      ...      ...
95     False  False  False  False      False  False      False
96     False  False  False  False      False  False      False
97     False  False  False  False      False  False      False
98     False  False  False  False      False  False      False
99     False  False  False  False      False  False      False

```

```

  Monthly_Stipend
0      False
1      False
2      False
3      False
4      False
..      ...
95     False
96     False
97     False
98     False
99     False

```

[100 rows x 8 columns]

```

--- Missing Values Count ---
Student_ID      0
Name            0
Age             6
Marks           6
Department      4
City            4
Attendance      5
Monthly_Stipend 4
dtype: int64

```

```

--- DataFrame After Removing Missing Rows ---
**
   Student_ID  Name  Age  Marks  Department  City  Attendance  \
0      1001  Tejaswi  18.0  90.0         IT    Hyderabad    92.21
1      1002    Anu   23.0  55.0        CIVIL    Tirupati    99.01
4      1005   Kiran   21.0  88.0         IT  Visakhapatnam    65.49
6      1007  Arjun   18.0  87.0        MECH     Guntur    69.61
8      1009  Manoj   19.0  70.0         ECE     Guntur    69.26
..      ...     ...   ...   ...         ...     ...     ...
95     1096  Harsha   20.0  49.0         EEE     Nellore    96.59
96     1097  Swathi   19.0  63.0        CIVIL  Vijayawada    99.28
97     1098   Varun   22.0  51.0         IT     Guntur    93.07
98     1099  Bhavya   22.0  63.0         ECE     Guntur    92.28
99     1100  Sandeep  18.0  97.0         CSE     Guntur    87.49

   Monthly_Stipend
0      17326.0
1      20580.0
4      21521.0
6      11461.0
8      24340.0
..      ...
95     10067.0
96     19437.0
97     16802.0
98      6002.0
99     6901.0

[71 rows x 8 columns]

```

```

...
--- DataFrame After Filling Missing Values ---
   Student_ID  Name  Age  Marks  Department  City  \
0      1001  Tejaswi  18.000000  90.0         IT    Hyderabad
1      1002    Anu   23.000000  55.0        CIVIL    Tirupati
2      1003   Priya  22.000000  89.0         IT    Vijayawada
3      1004    Ravi  21.191489  45.0        MECH     Nellore
4      1005   Kiran  21.000000  88.0         IT  Visakhapatnam
..      ...     ...   ...   ...         ...     ...
95     1096  Harsha  20.000000  49.0         EEE     Nellore
96     1097  Swathi  19.000000  63.0        CIVIL  Vijayawada
97     1098   Varun  22.000000  51.0         IT     Guntur
98     1099  Bhavya  22.000000  63.0         ECE     Guntur
99     1100  Sandeep  18.000000  97.0         CSE     Guntur

   Attendance  Monthly_Stipend
0      92.21      17326.0
1      99.01      20580.0
2      NaN      19781.0
3      70.04      7691.0
4      65.49      21521.0
..      ...     ...
95     96.59      10067.0
96     99.28      19437.0
97     93.07      16802.0
98     92.28      6002.0
99     87.49      6901.0

[100 rows x 8 columns]

```

Task 8: Merge, Concatenate, GroupBy & Pivot Table

In this task, I performed data manipulation operations using Pandas, including merging and concatenating DataFrames, GroupBy analysis, and creating Pivot Tables. I combined datasets using a common column, joined rows from DataFrames, calculated department-wise statistics such as average marks, and summarized data using Pivot Tables. This task helped me understand how to combine and summarize data for analytical purposes.

Output:

TASK 8: MERGE, CONCATENATE, GROUPBY & PIVOT TABLE

Original Dataset:

	Student_ID	Name	Age	Marks	Department	City	Attendance	\
0	1001	Tejaswi	18	90	IT	Hyderabad	92.21	
1	1002	Anu	23	55	CIVIL	Tirupati	99.01	
2	1003	Priya	22	89	IT	Vijayawada	82.53	
3	1004	Ravi	21	45	MECH	Nellore	70.04	
4	1005	Kiran	21	88	IT	Visakhapatnam	65.49	

Monthly_Stipend

0	17326
1	20580
2	19781
3	7691
4	21521

--- 1. MERGE ---

	Student_ID	Name	Age	Marks	Department	City	Attendance	\
0	1001	Tejaswi	18	90	IT	Hyderabad	92.21	
1	1002	Anu	23	55	CIVIL	Tirupati	99.01	
2	1003	Priya	22	89	IT	Vijayawada	82.53	
3	1004	Ravi	21	45	MECH	Nellore	70.04	
4	1005	Kiran	21	88	IT	Visakhapatnam	65.49	

Monthly_Stipend Scholarship_Status Club

0	17326	Yes	Coding
1	20580	No	Sports
2	19781	Yes	Coding
3	7691	No	Cultural
4	21521	Yes	Sports

--- 2. CONCATENATE ---

	Student_ID	Name	Age	Marks	Department	City	Attendance	\
0	1001	Tejaswi	18	90	IT	Hyderabad	92.21	
1	1002	Anu	23	55	CIVIL	Tirupati	99.01	
2	1003	Priya	22	89	IT	Vijayawada	82.53	
3	1004	Ravi	21	45	MECH	Nellore	70.04	
4	1005	Kiran	21	88	IT	Visakhapatnam	65.49	
5	1006	Sneha	24	88	ECE	Vijayawada	73.04	
6	1007	Arjun	18	87	MECH	Guntur	69.61	
7	1008	Divya	22	81	CIVIL	Visakhapatnam	88.72	
8	1009	Manoj	19	70	ECE	Guntur	69.26	
9	1010	Lakshmi	18	83	MECH	Vijayawada	82.72	

Monthly_Stipend

0	17326
1	20580
2	19781
3	7691
4	21521
5	15721
6	11461
7	15284
8	24340
9	22152

--- 3. GROUPBY ---

	Student_Count	Average_Marks	Minimum_Marks	Maximum_Marks	\
Department					
CIVIL	10	69.00	51	84	
CSE	16	70.25	47	99	
ECE	19	73.84	50	96	
EEE	16	72.19	46	99	
IT	21	70.29	46	90	
MECH	18	72.17	45	99	

Department	Average_Attendance
CIVIL	87.55
CSE	81.13
ECE	78.58
EEE	81.35
IT	82.31
MECH	80.88

--- 4. PIVOT TABLE ---

Department	Attendance	Marks
CIVIL	87.55	69.00
CSE	81.13	70.25
ECE	78.58	73.84
EEE	81.35	72.19
IT	82.31	70.29
MECH	80.88	72.17

Task 9: Exporting Data

In this task, I exported the processed DataFrame into a new CSV file using Pandas. After saving the data, I loaded the exported file again to verify that the processed information was stored correctly. This task helped me understand how to save, share, and verify analytical datasets for future use and project submission.

Output:

... ORIGINAL DATASET

	Student_ID	Name	Age	Marks	Department	City	Attendance	\
0	1001	Tejaswi	18	90	IT	Hyderabad	92.21	
1	1002	Anu	23	55	CIVIL	Tirupati	99.01	
2	1003	Priya	22	89	IT	Vijayawada	82.53	
3	1004	Ravi	21	45	MECH	Nellore	70.04	
4	1005	Kiran	21	88	IT	Visakhapatnam	65.49	

Monthly_Stipend

0	17326
1	20580
2	19781
3	7691
4	21521

PROCESSED DATASET

	Student_ID	Name	Age	Marks	Department	City	Attendance	\
44	1045	Kiran	18	99	EEE	Nellore	86.21	
60	1061	Tejaswi	24	99	MECH	Nellore	70.12	
40	1041	Tejaswi	19	99	CSE	Visakhapatnam	94.03	
94	1095	Keerthi	24	97	EEE	Tirupati	94.53	
99	1100	Sandeep	18	97	CSE	Guntur	87.49	

Monthly_Stipend

44	20406
60	8534
40	18417
94	6802
99	6901

File exported successfully!

EXPORTED FILE DATA

	Student_ID	Name	Age	Marks	Department	City	Attendance \
0	1045	Kiran	18	99	EEE	Nellore	86.21
1	1061	Tejaswi	24	99	MECH	Nellore	70.12
2	1041	Tejaswi	19	99	CSE	Visakhapatnam	94.03
3	1095	Keerthi	24	97	EEE	Tirupati	94.53
4	1100	Sandeep	18	97	CSE	Guntur	87.49

Monthly_Stipend

0	20406
1	8534
2	18417
3	6802
4	6901

Rows: 100

Columns: 8

TASK 9 COMPLETED SUCCESSFULLY!

Task 10: Mini Data Analysis Project

In this task, I performed a mini data analysis project using Python, Pandas, and NumPy on a student performance dataset. I loaded and inspected the dataset, checked and handled missing values, filtered and sorted student records, performed department-wise GroupBy analysis, and created a Pivot Table. I identified useful insights, including the highest marks, students scoring 80 or above, average marks by department, and pass percentage based on a defined threshold. Finally, I exported the processed dataset into a CSV file. This project helped me apply fundamental data analysis techniques to a real-world-style dataset and understand how to derive meaningful insights from data.

```
=====
... MINI DATA ANALYSIS PROJECT
=====

--- 1. LOADING DATASET ---
Student_ID  Name  Age  Marks  Department  City  Attendance \
0      1001  Tejaswi  18    90         IT    Hyderabad    92.21
1      1002    Anu   23    55        CIVIL    Tirupati    99.01
2      1003   Priya  22    89         IT    Vijayawada    82.53
3      1004    Ravi  21    45        MECH    Nellore     70.04
4      1005   Kiran  21    88         IT    Visakhapatnam    65.49

Monthly_Stipend
0      17326
1      20580
2      19781
3       7691
4      21521

--- 2. DATA INSPECTION ---

Dataset Shape:
(100, 8)

Data Types:
Student_ID      int64
Name            object
Age             int64
Marks           int64
Department      object
City            object
Attendance      float64
Monthly_Stipend int64
dtype: object
```

Output:

Statistical Summary:

```

***
count    Student_ID    Age    Marks    Attendance    Monthly_Stipend
mean    100.000000    100.000000    100.00000    100.000000    100.000000
std      29.011492     1.910603     15.24378    10.731613     5540.901886
min      1001.000000    18.000000    45.00000    65.490000     5702.000000
25%      1025.750000    20.000000    60.00000    70.687500     11330.000000
50%      1050.500000    21.000000    72.00000    81.730000     15891.000000
75%      1075.250000    23.000000    83.25000    92.227500     20449.500000
max      1100.000000    24.000000    99.00000    99.280000     24934.000000

```

--- 3. MISSING VALUES ---

```

Student_ID    0
Name          0
Age           0
Marks         0
Department    0
City          0
Attendance    0
Monthly_Stipend    0
dtype: int64

```

--- 4. HANDLE MISSING VALUES ---

Missing values handled successfully!

--- 5. FILTERING DATA ---

*** Students with Marks >= 80:

```

Student_ID    Name    Marks    Department
0            1001    Tejaswi    90        IT
2            1003    Priya      89        IT
4            1005    Kiran      88        IT
5            1006    Sneha      88        ECE
6            1007    Arjun      87        MECH
7            1008    Divya      81        CIVIL
9            1010    Lakshmi    83        MECH
11           1012    Pooja      87        MECH
21           1022    Anu        81        IT
22           1023    Priya      80        MECH
24           1025    Kiran      92        MECH
27           1028    Divya      87        EEE
34           1035    Keerthi    88        IT
40           1041    Tejaswi    99        CSE
44           1045    Kiran      99        EEE
45           1046    Sneha      91        CSE
48           1049    Manoj      90        MECH
50           1051    Sai        92        ECE
52           1053    Rahul      95        ECE
55           1056    Harsha     81        CSE
59           1060    Sandeep    88        EEE
60           1061    Tejaswi    99        MECH
61           1062    Anu        81        ECE
65           1066    Sneha      89        EEE
72           1073    Rahul      87        CSE
73           1074    Nikhil     84        CIVIL
74           1075    Keerthi    83        IT
76           1077    Swathi     84        IT
78           1079    Bhavya     94        CSE
80           1081    Tejaswi    96        ECE
83           1084    Ravi       83        IT
90           1091    Sai        82        MECH
94           1095    Keerthi    97        EEE
99           1100    Sandeep    97        CSE

```

```

--- 6. SORTING DATA ---
TOP 10 STUDENTS:
...
  Student_ID  Name  Marks  Department
44      1045   Kiran    99         EEE
60      1061  Tejaswi   99         MECH
40      1041  Tejaswi   99         CSE
94      1095  Keerthi   97         EEE
99      1100  Sandeep   97         CSE
80      1081  Tejaswi   96         ECE
52      1053   Rahul   95         ECE
78      1079  Bhavya   94         CSE
24      1025   Kiran   92         MECH
50      1051    Sai    92         ECE

--- 7. GROUPBY ANALYSIS ---
      Students  Average_Marks  Average_Attendance  Total_Stipend
Department
CIVIL           10           69.00           87.55         152191
CSE             16           70.25           81.13         249259
ECE             19           73.84           78.58         306813
EEE             16           72.19           81.35         243877
IT              21           70.29           82.31         333824
MECH            18           72.17           80.88         286523

--- 8. PIVOT TABLE ---
      Attendance  Marks
Department
CIVIL           87.55  69.00
CSE             81.13  70.25
ECE             78.58  73.84
EEE             81.35  72.19
IT              82.31  70.29
MECH            80.88  72.17

--- 9. USEFUL INSIGHTS ---
Highest Marks: 99
Student Name: Tejaswi
Students Scoring 80 or Above: 34
Pass Percentage: 100.0 %
Department with Highest Average Marks: ECE

--- 10. EXPORTING DATA ---
Processed dataset exported successfully!

=====
TASK 10 COMPLETED SUCCESSFULLY!
=====

```

Task 10: Key Insights

- Highest Marks:** The highest marks in the dataset are 99, achieved by multiple students. This insight was identified using the `max()` function to find the highest score.
- High-Scoring Students:** A total of 34 students scored 80 or above. This was identified by filtering the dataset using the condition `Marks >= 80`.
- Department-wise Performance:** ECE had the highest average marks of 73.84 among the departments. This was calculated using GroupBy analysis and the mean function.
- Pass Percentage:** The dataset shows a 100% pass percentage based on the defined threshold of 40 marks, as all 100 students scored at least 40.
- Attendance Analysis:** CIVIL had the highest average attendance of 87.55% among the departments. This was identified by calculating the average attendance using GroupBy analysis.